Chittagong University of Engineering & Technology

Department of Electronics and Telecommunication Engineering

EXPERIMENT NO: 01

Name of the Experiment

Configuration of PC to PC Network Using Crossover Cable

Course No : ETE 452

Course Title : Advanced Communications and Networking Sessional

Date of Experiment : 11/01/2026

Date of Submission : 25/01/2026

REMARKS

Name : Showkot Hosen

Roll no : 2008010

Level :L-4

Term :T-II

Objectives

- To establish direct communication between two PCs using a crossover cable.
- To verify data transmission without using any intermediate networking device.

Software

- Cisco Packet Tracer

Procedure

1. Two PCs were placed in the workspace of Cisco Packet Tracer.
2. A crossover cable was selected and connected between the FastEthernet ports of both PCs.
3. IP addresses were manually assigned to both PCs within the same network.
4. Network connectivity was tested using the ping command.

Connection Diagram

Figure 1 : PC to PC connection using crossover wire in CISCO Packet Tracer

Discussion

1. Direct communication between two PCs was successfully established using a crossover cable.
2. It was observed that no hub or switch was required for data transmission.
3. Both PCs were required to be in the same IP network for proper communication.
4. Successful ping replies confirmed correct physical and logical configuration.
5. This experiment demonstrated the basic concept of peer-to-peer networking.

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EXPERIMENT NO: 02

Name of the Experiment

Implementing Local Area Network Using Hub and Switch

Course No : ETE 452

Course Title : Advanced Communications and Networking Sessional

Date of Experiment : 11/01/2026

Date of Submission : 25/01/2026

REMARKS

Name : Showkot Hosen

Roll no : 2008010

Level :L-4

Term :T-II

Objectives

- To implement a LAN using a hub and a switch.
- To observe data communication among multiple PCs in a LAN.

Software

- Cisco Packet Tracer

Introduction

A Local Area Network (LAN) connects multiple devices within a limited area.
A hub broadcasts data to all ports, whereas a switch forwards data to specific devices.
LAN improves resource sharing and communication efficiency.

Procedure

1. Multiple PCs were added to the workspace.
2. A hub and a switch were placed separately for implementation.
3. Straight-through cables were used to connect PCs to the hub and switch.
4. IP addresses were assigned to all PCs in the same subnet.
5. Connectivity was verified using the ping command.

Connection Diagram

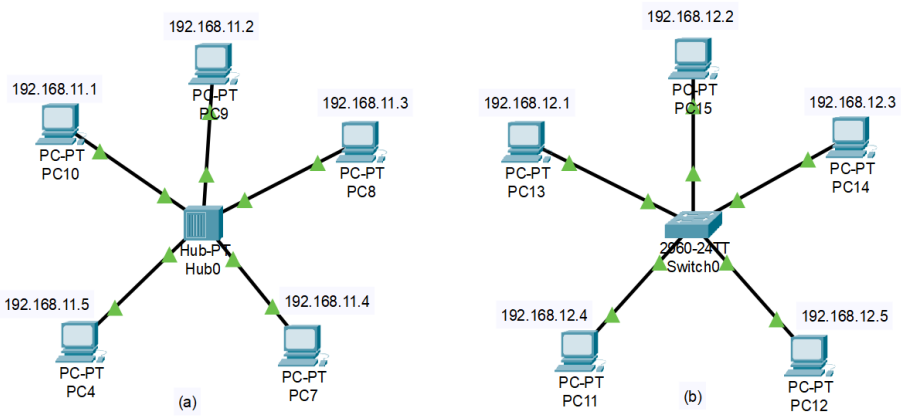

Figure 2 : Local Area Network (LAN) using (a) Hub and (b) Switch

A switch is considered smarter than a hub because MAC address learning and forwarding decisions are performed using a MAC address table. Frames are forwarded only to the specific destination port. In contrast, a hub does not perform MAC address resolution and blindly broadcasts data to all connected ports

Discussion

1. LAN connectivity was successfully implemented using both hub and switch.
2. It was observed that the hub broadcasted data to all connected PCs.
3. The switch transmitted data only to the intended destination PC.
4. Reduced network traffic was noticed in the switch-based LAN.
5. This experiment highlighted the performance difference between hub and switch.

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EXPERIMENT NO: 03

Name of the Experiment

Configuration of Routers as Default Gateway in LAN

Course No : ETE 452

Course Title : Advanced Communications and Networking Sessional

Date of Experiment : 11/01/2026

Date of Submission : 25/01/2026

REMARKS

Name : Showkot Hosen

Roll no : 2008010

Level :L-4

Term :T-II

Objectives

- To configure a router for inter-network communication.
- To understand the role of default gateway in a LAN.

Software

- Cisco Packet Tracer

Introduction

A router is used to connect multiple networks together.The default gateway allows devices to communicate with external networks.Routing enables packet forwarding between different IP networks.

Procedure

1. Multiple PCs, a switch, and a router were placed in the workspace.
2. PCs were connected to the switch using straight-through cables.
3. The switch was connected to the router interface.
4. IP addresses were assigned to PCs and router interfaces.
5. The router IP was set as the default gateway for all PCs.
6. Connectivity was tested using ping across networks.

Connection Diagram

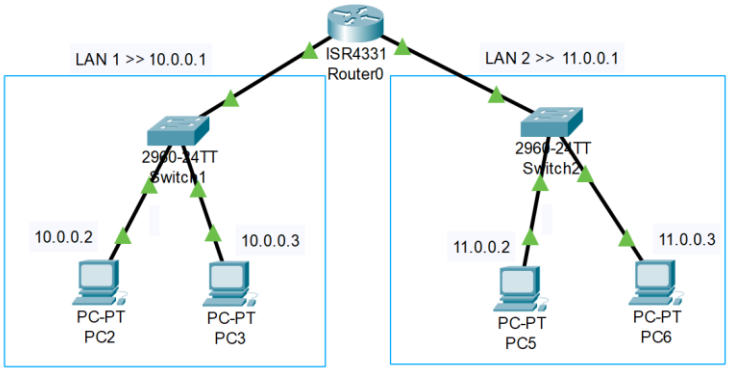

Figure 3 : Local Area Network (LAN) using (a) Hub and (b) Switch

Discussion

1. Successful communication between different networks was achieved using a router.
2. The importance of default gateway configuration was clearly observed.
3. Without a default gateway, inter-network communication was not possible.
4. Router interfaces were required to be properly configured and enabled.
5. This experiment demonstrated the basic concept of routing in a LAN environment.